In [1]:
```
# Cell 1: imports & helper
import os
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# make output folders
os.makedirs("outputs", exist_ok=True)
os.makedirs("data", exist_ok=True)

# helper to interpret boolean-like columns reliably
def truthy(series):
    # convert to str then check common true values
    return series.astype(str).str.lower().isin(['true','t','1','yes','y'])
```

In [2]:
```
# Cell 2: load CSV (adjust path if different)
csv_path = "data/squirrels.csv"
df = pd.read_csv(csv_path)
df.shape, df.columns.tolist()
```

Out[2]:
```
((3023, 31),
 ['X',
  'Y',
  'Unique Squirrel ID',
  'Hectare',
  'Shift',
  'Date',
  'Hectare Squirrel Number',
  'Age',
  'Primary Fur Color',
  'Highlight Fur Color',
  'Combination of Primary and Highlight Color',
  'Color notes',
  'Location',
  'Above Ground Sighter Measurement',
  'Specific Location',
  'Running',
  'Chasing',
  'Climbing',
  'Eating',
  'Foraging',
  'Other Activities',
  'Kuks',
  'Quaas',
  'Moans',
  'Tail flags',
  'Tail twitches',
  'Approaches',
  'Indifferent',
  'Runs from',
  'Other Interactions',
  'Lat/Long'])
```

In [3]:
```python
# Cell 3: inspect & basic cleaning
df = df.drop_duplicates()   # drop exact duplicate rows if any
df.columns = [c.strip() for c in df.columns]  # trim whitespace
df.info()
df.head(5)
```

```
<class 'pandas.DataFrame'>
RangeIndex: 3023 entries, 0 to 3022
Data columns (total 31 columns):
 #   Column                                      Non-Null Count  Dtype  
---  ------                                      --------------  -----  
 0   X                                           3023 non-null   float64
 1   Y                                           3023 non-null   float64
 2   Unique Squirrel ID                          3023 non-null   str    
 3   Hectare                                     3023 non-null   str    
 4   Shift                                       3023 non-null   str    
 5   Date                                        3023 non-null   int64  
 6   Hectare Squirrel Number                     3023 non-null   int64  
 7   Age                                         2902 non-null   str    
 8   Primary Fur Color                           2968 non-null   str    
 9   Highlight Fur Color                         1937 non-null   str    
 10  Combination of Primary and Highlight Color  3023 non-null   str    
 11  Color notes                                 182 non-null    str    
 12  Location                                    2959 non-null   str    
 13  Above Ground Sighter Measurement            2909 non-null   str    
 14  Specific Location                           476 non-null    str    
 15  Running                                     3023 non-null   bool   
 16  Chasing                                     3023 non-null   bool   
 17  Climbing                                    3023 non-null   bool   
 18  Eating                                      3023 non-null   bool   
 19  Foraging                                    3023 non-null   bool   
 20  Other Activities                            437 non-null    str    
 21  Kuks                                        3023 non-null   bool   
 22  Quaas                                       3023 non-null   bool   
 23  Moans                                       3023 non-null   bool   
 24  Tail flags                                  3023 non-null   bool   
 25  Tail twitches                               3023 non-null   bool   
 26  Approaches                                  3023 non-null   bool   
 27  Indifferent                                 3023 non-null   bool   
 28  Runs from                                   3023 non-null   bool   
 29  Other Interactions                          240 non-null    str    
 30  Lat/Long                                    3023 non-null   str    
dtypes: bool(13), float64(2), int64(2), str(14)
memory usage: 463.6 KB
```

Out[3]:

	X	Y	Unique Squirrel ID	Hectare	Shift	Date	Hectare Squirrel Number	Age	Primary Fur Color	Hig Fur
0	-73.956134	40.794082	37F-PM-1014-03	37F	PM	10142018	3	NaN	NaN	
1	-73.968857	40.783783	21B-AM-1019-04	21B	AM	10192018	4	NaN	NaN	
2	-73.974281	40.775534	11B-PM-1014-08	11B	PM	10142018	8	NaN	Gray	
3	-73.959641	40.790313	32E-PM-1017-14	32E	PM	10172018	14	Adult	Gray	
4	-73.970268	40.776213	13E-AM-1017-05	13E	AM	10172018	5	Adult	Gray	Cinı

5 rows × 31 columns

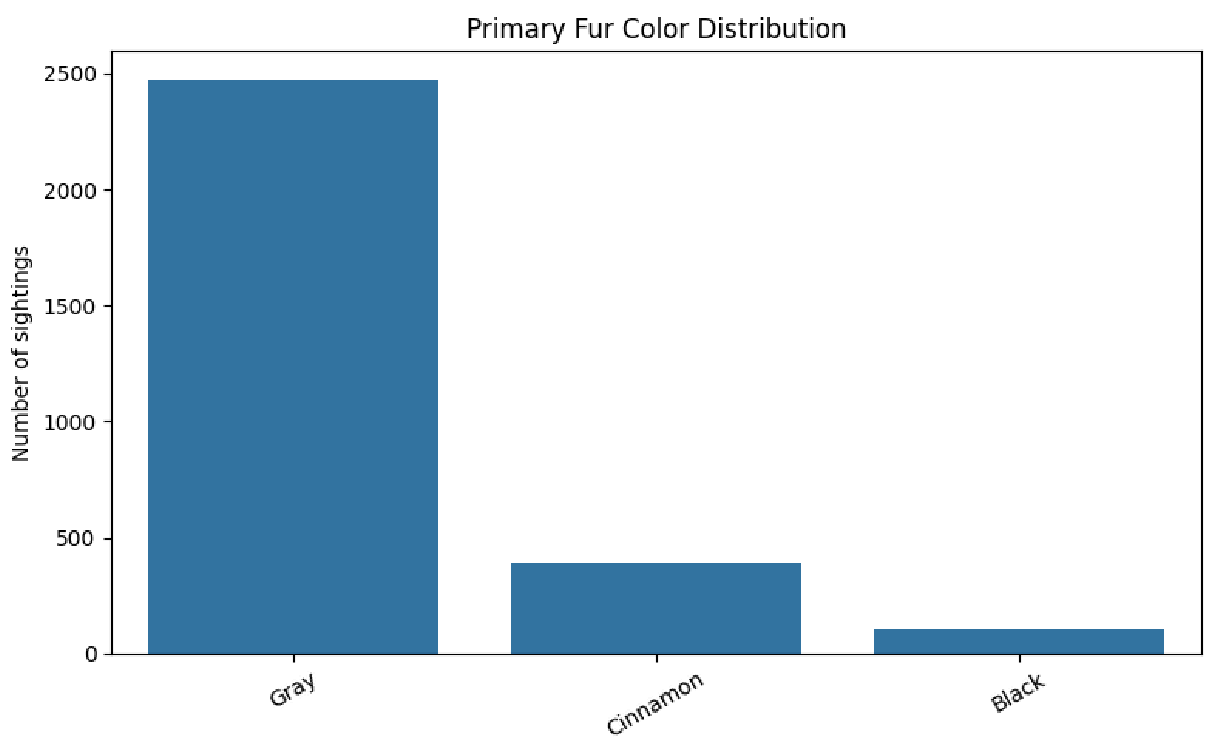
In [4]:
```
# Cell 4: make boolean columns robust
for col in ['Running','Chasing','Climbing','Eating','Foraging','Approaches','Indiff
    if col in df.columns:
        df[col + '_bool'] = truthy(df[col])
# Example: Approaches_bool will be True/False
```

In [5]:
```
# Cell 5: summary stats
print("Total sightings:", len(df))

if 'Primary Fur Color' in df.columns:
    color_counts = df['Primary Fur Color'].fillna("Unknown").value_counts()
    print("\nFur color counts:\n", color_counts)

# approach rate
if 'Approaches_bool' in df.columns:
    approach_rate = df['Approaches_bool'].mean()
    print(f"\nApproach rate: {approach_rate:.2%} (fraction of sightings where squir
```

```
Total sightings: 3023

Fur color counts:
 Primary Fur Color
Gray        2473
Cinnamon     392
Black        103
Unknown       55
Name: count, dtype: int64

Approach rate: 5.89% (fraction of sightings where squirrel approached a human)
```

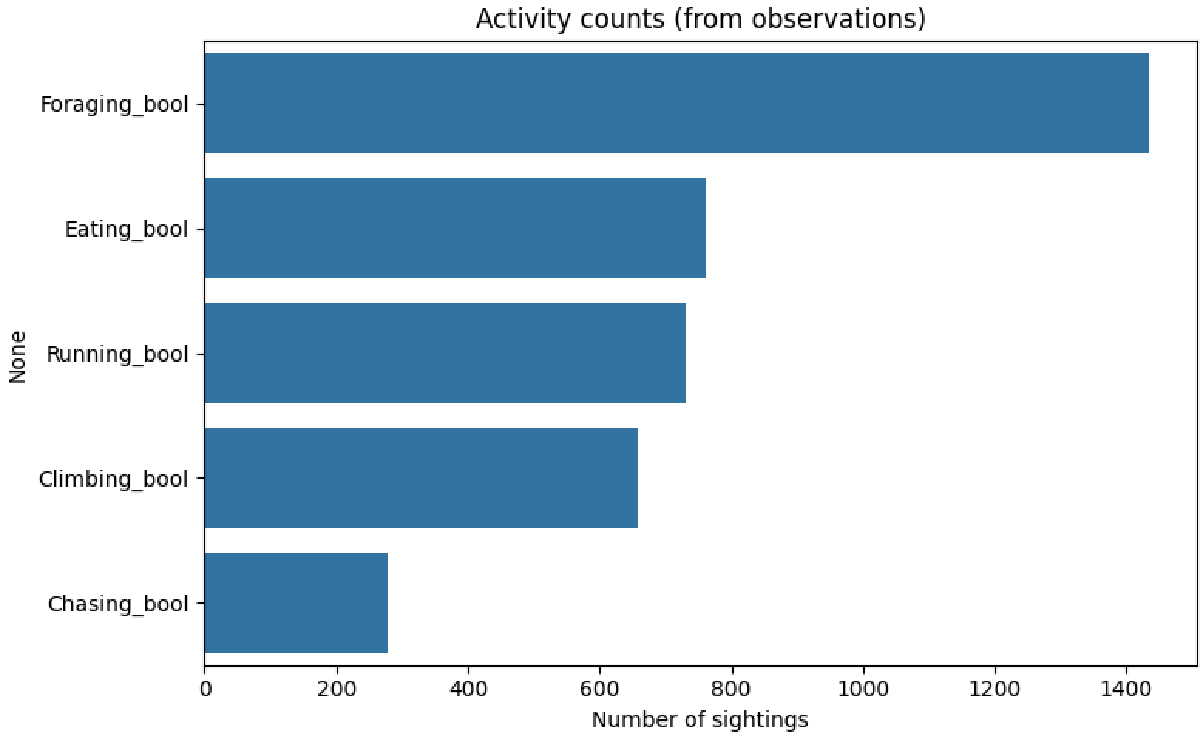
In [6]:

```
# Cell 6: bar chart of fur color
plt.figure(figsize=(8,5))
sns.countplot(data=df, x='Primary Fur Color', order=df['Primary Fur Color'].value_c
plt.title('Primary Fur Color Distribution')
plt.xlabel('')
plt.ylabel('Number of sightings')
plt.xticks(rotation=30)
plt.tight_layout()
plt.savefig('outputs/fur_color.png', dpi=300, bbox_inches='tight')
plt.show()
```



In [7]:

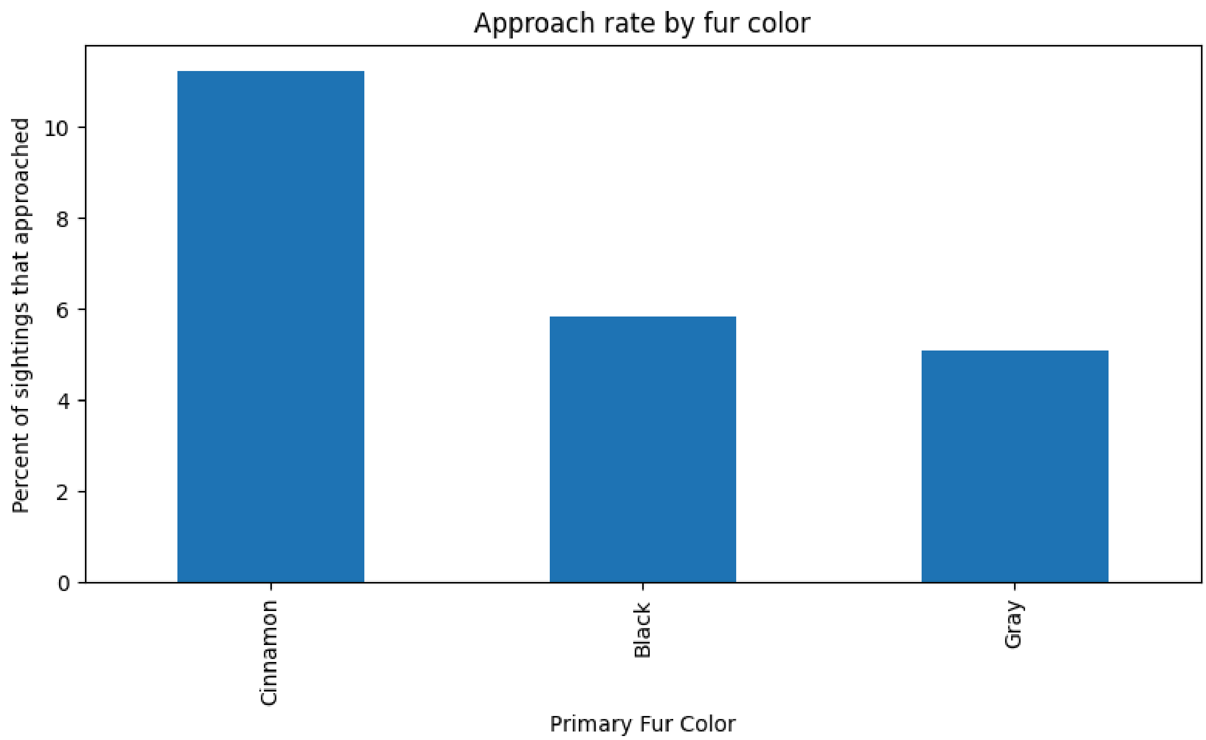
```
# Cell 7: activity counts
activity_cols = [c + '_bool' for c in ['Running','Chasing','Climbing','Eating','For
activity_counts = df[activity_cols].sum().sort_values(ascending=False)
plt.figure(figsize=(8,5))
sns.barplot(x=activity_counts.values, y=activity_counts.index)
plt.title('Activity counts (from observations)')
plt.xlabel('Number of sightings')
plt.tight_layout()
plt.savefig('outputs/activity_counts.png', dpi=300, bbox_inches='tight')
plt.show()
```



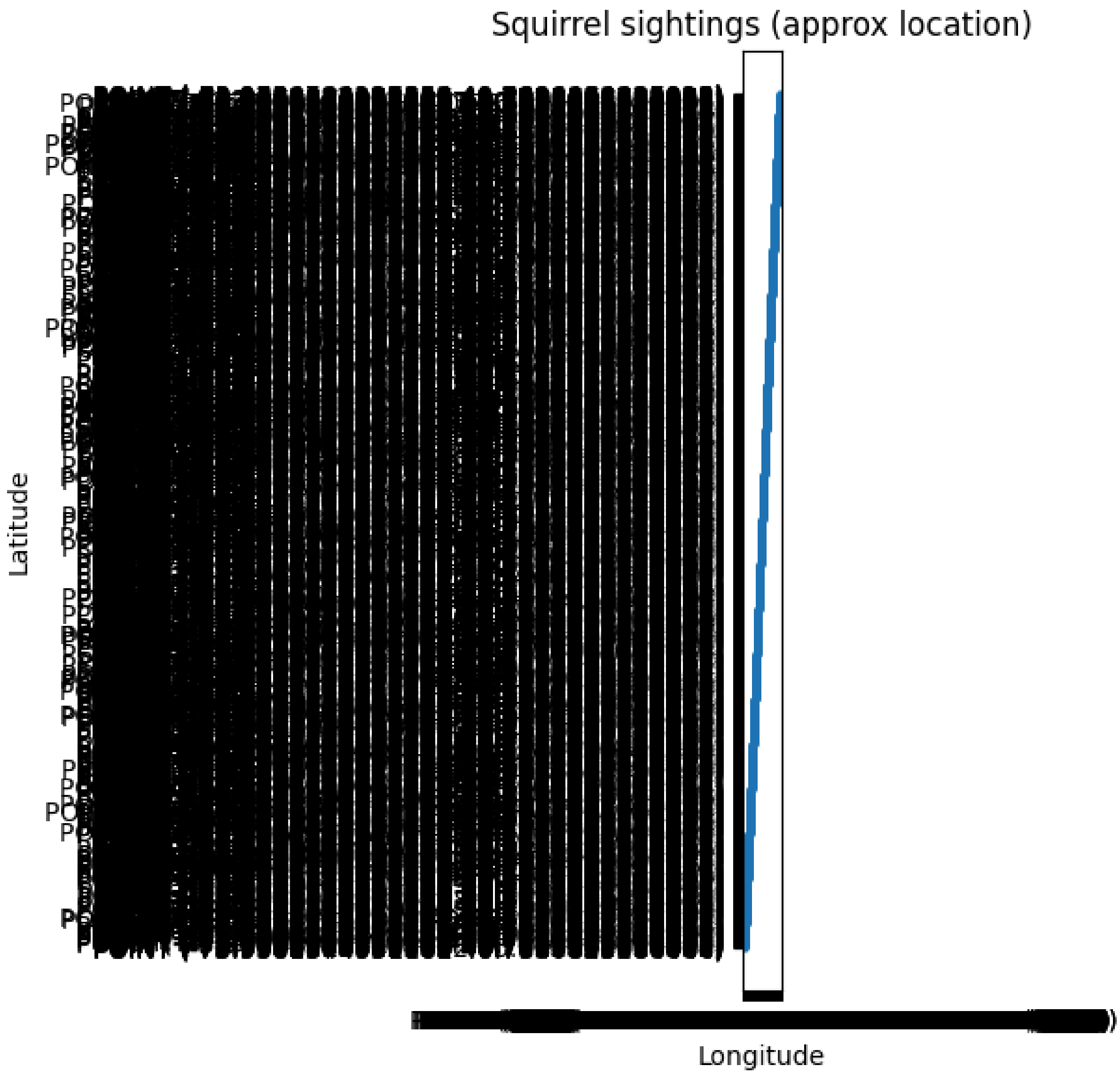
In [8]:

```
# Cell 8: approach by color
if 'Approaches_bool' in df.columns and 'Primary Fur Color' in df.columns:
    approaches_by_color = df[df['Approaches_bool']].groupby('Primary Fur Color').si
    total_by_color = df.groupby('Primary Fur Color').size()
    rate_by_color = (approaches_by_color / total_by_color).fillna(0).sort_values(as
    rate_by_color = rate_by_color * 100  # percent
    print("Approach rate by primary fur color (percent):\n", rate_by_color.round(2)

    plt.figure(figsize=(8,5))
    rate_by_color.plot(kind='bar')
    plt.ylabel('Percent of sightings that approached')
    plt.title('Approach rate by fur color')
    plt.tight_layout()
    plt.savefig('outputs/approach_by_color.png', dpi=300, bbox_inches='tight')
    plt.show()
```

```
Approach rate by primary fur color (percent):
 Primary Fur Color
Cinnamon    11.22
Black        5.83
Gray         5.10
dtype: float64
```



In [9]:
```
# Cell 9: map scatter (latitude/longitude)
lat_cols = [c for c in df.columns if 'lat' in c.lower()]
lon_cols = [c for c in df.columns if 'long' in c.lower() or 'lon' in c.lower()]

if lat_cols and lon_cols:
    lat_col = lat_cols[0]
    lon_col = lon_cols[0]
    plt.figure(figsize=(6,6))
    plt.scatter(df[lon_col], df[lat_col], s=4, alpha=0.6)
    plt.title('Squirrel sightings (approx location)')
    plt.xlabel('Longitude'); plt.ylabel('Latitude')
    plt.tight_layout()
    plt.savefig('outputs/sightings_map.png', dpi=300, bbox_inches='tight')
    plt.show()
else:
    print("No lat/long columns detected. (OK to skip)")
```



In [10]:
```
# Cell 10: final printed summary
approach_percent = (df['Approaches_bool'].mean() * 100) if 'Approaches_bool' in df.
print("=== SUMMARY ===")
print(f"Total sightings: {len(df)}")
if approach_percent is not None:
    print(f"Approx approach rate: {approach_percent:.2f}%")
if 'Primary Fur Color' in df.columns:
    top_colors = df['Primary Fur Color'].value_counts().head(3)
    print("Top 3 fur colors:\n", top_colors.to_string())
```

```
=== SUMMARY ===
Total sightings: 3023
Approx approach rate: 5.89%
Top 3 fur colors:
 Primary Fur Color
Gray        2473
Cinnamon     392
Black        103
```